University Campus Network Design: Technical Report

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1. Introduction & Executive Summary

This technical report outlines the comprehensive network planning, topology design, and configuration for the University campus. The enterprise network spans two geographically separated campuses located 20 miles apart, integrating four academic faculties, administrative divisions, student computer labs, and central IT infrastructure. The primary objective is to deliver secure, reliable, and high-performance interconnectivity adhering to modern hierarchical network standards.

2. Network Topology & Architecture

The network design implements a modular hierarchical model comprising Access, Distribution, and Core layers across the main and remote campuses:

- **Main Campus:** Comprises Buildings A, B, and C. Building A houses administrative staff (Management, HR, Finance) and the Faculty of Business, utilizing VLAN segmentation and centralized DHCP. Building B accommodates Engineering, Computing, and Art/Design labs. Building C hosts core IT services, including the University Web server.
- **Smaller Campus (Remote Site):** Located 20 miles away, housing the Faculty of Health and Sciences. Staff offices and student labs are separated across distinct physical floors, connected via wide-area serial routing.
- **External Cloud Services:** Integrates an external cloud-hosted email server accessible via static routing and WAN links.

3. Addressing Scheme & VLAN Plan

To isolate broadcast domains and enforce security, logical segmentation is applied across the infrastructure:

VLAN ID	Department / Area	Network Address	Subnet Mask	Default Gateway
10	Administration (Mgmt/HR/Finance)	192.168.10.0	255.255.255.0	192.168.10.1
20	Faculty of Business	192.168.20.0	255.255.255.0	192.168.20.1
30	Engineering, Computing & Art/Design	192.168.30.0	255.255.255.0	192.168.30.1
40	Student Labs & IT Services	192.168.40.0	255.255.255.0	192.168.40.1
50	Health and Sciences (Remote Campus)	192.168.50.0	255.255.255.0	192.168.50.1

4. Routing & Services Configuration

Dynamic and static routing mechanisms ensure seamless end-to-end packet delivery:

- **Inter-VLAN Routing:** Configured via Router-on-a-Stick sub-interfaces on the core gateway router using 802.1Q encapsulation.
- **Dynamic Routing (RIPv2):** Deployed across internal routers to exchange routing updates dynamically, ensuring fault tolerance and automated path selection.
- **Static Routing:** Implemented at the network edge to forward external traffic toward the cloud-hosted email server.
- **DHCP Implementation:** A router-based DHCP server automatically assigns IP parameters, subnet masks, and default gateways to workstations in Building A.

5. Security & Access Control

Security is enforced at multiple layers within the network topology:

- **Port Security:** Configured on access layer switches to restrict unauthorized physical device attachments by limiting maximum MAC addresses and enabling sticky learning.
- **VLAN Isolation:** Administrative traffic is logically separated from student and guest labs to prevent unauthorized access to sensitive financial and management records.

6. Critical Evaluation & Reflection

Performance & Scalability: The hierarchical structure minimizes broadcast domain overhead and optimizes traffic flow. The modular IP addressing scheme allows straightforward expansion for future faculty additions without address re-allocation.

Reliability & Redundancy: While RIPv2 provides basic internal path recovery, future iterations could integrate OSPF and Spanning Tree Protocol (STP) enhancements (such as Rapid PVST+ and EtherChannel link aggregation) to eliminate single points of failure and reduce convergence times.

Self-Reflection: Simulating this campus network in Cisco Packet Tracer provided practical insights into real-world enterprise deployment challenges, emphasizing the importance of rigorous IP planning and proactive security configurations.

7. References

Cisco Systems. (2020). *Enterprise Campus Architecture Overview*. Cisco Press.
Odom, W. (2019). *CCNA 200-301 Official Cert Guide*. Cisco Press.